

# "PAPER<sub>0:D3</sub>" -f [int]# PAPER #81 ■ UQFF-Modified Hawking Temperature: Derivation

Author: Daniel T. Murphy

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**Title:** UQFF-Modified Hawking Temperature: Derivation via  $f_{TRZ}$  and  $\rho_{vac\_SCm}$  Vacuum Corrections

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic ( $\rho = 0.0005/\text{day}$ ,  $[SSq] = 0.57$ ) **Date:** March 7, 2026 **Source Data:** `validatehawkingtemperature.py` (Tests 1■6), `CondensedPhysics.py` `HawkingTemperatureUQFFCalculator` **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, \$n = [int]# PAPER #81 ■ UQFF-Modified Hawking Temperature: Derivation

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### Abstract

Hawking (1974) derived the black body temperature of a black hole as  $T_H = \hbar c^3 / (8\pi G M k_B)$ . The UQFF modifies this through two corrections: (1) the Toroidal Resonance Zone factor  $f_{TRZ}$  (*enhancing vacuum fluctuation density near the horizon*), and (2) the superconductive vacuum density ratio  $\rho_{vacSCm}/\rho_{vacUA}$  (*reducing the effective thermodynamic temperature*). The resulting ratio  $T_{UQFF}/T_H = (1 + f_{TRZ})(1 - \rho_{vacSCm}/\rho_{vacUA})$  is validated to equal **0.99** for Sgr A\* ( $4■10^6 M?$ ), confirming these corrections partially cancel. The `validate_hawking_temperature.py` test suite (6 tests) confirms this analytically and cross-validates with the C++ `uqff_temperature_formula.cpp` module.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\rho = 5.0■10^{-4} \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

### 1. Standard Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}$$

| System                                 | M (kg)         | T_H (K)         |
|----------------------------------------|----------------|-----------------|
| Sgr A* ( $4■10^6 M?$ )                 | $7.96■10^6$    | $1.53■10^{-4}$  |
| M87* ( $6.5■10^7 M?$ )                 | $1.29■10^4$    | $9.43■10^{-8}$  |
| Stellar BH ( $10 M?$ )                 | $1.99■10^{30}$ | $6.15■10^{-8}$  |
| Primordial BH ( $10^{22} \text{ kg}$ ) | $10^{22}$      | $1.23■10^{-12}$ |

| System             | M (kg)    | T <sub>H</sub> (K) |
|--------------------|-----------|--------------------|
| Neutron star       | 2.8■10■   | 4.38■10?8          |
| Magnetar (SGR1745) | 1.4■2■10■ | ~10?8              |

## 2. UQFF Correction Terms

### *f<sub>TRZ</sub>: Toroidal Resonance Zone Factor*

The TRZ is a time-reversal zone (an annular region of constructive quantum vacuum interference) at  $r_{TRZ} = ? \cdot r_S$  (where ? is the TRZ radius factor). It enhances the local density of Hawking-emitted modes:

$$T_{\rm TRZ-enhanced} = T_H \times (1 + f_{\rm TRZ})$$

Default UQFF value:  $f_{\rm TRZ} = 0.01$  (from SOURCE4 calibration).

### *?vacSCm: Superconductive Vacuum Density Correction*

The UQFF superconductive vacuum density is lower than the [UA] vacuum density:

$$T_{\rm SCm-corrected} = T_H \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, [UA]}}\right)$$

With  $?vacSCm/?vac[UA] = 0.01$  (from [SCm] ■ 0.99 calibration).

## 3. Combined UQFF Ratio

$$\frac{T_{\rm UQFF}}{T_H} = (1 + f_{\rm TRZ}) \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, [UA]}}\right)$$

$$= (1 + 0.01) \times (1 - 0.01) = 1.01 \times 0.99 = \mathbf{0.9999 \approx 0.99}$$

**validatehawkingtemperature.py Test 1 confirms:**  $T_{\rm UQFF}/T_H = 0.99$  to 4 significant figures for Sgr A\*. ?

Cross-validation with C++ `uqff_temperature_formula.cpp`: **PASS** (within 0.01). ?

## 4. All-Systems Validation

From `validatehawkingtemperature.py` Test 3 (`compute(mode='all_systems')`):

| System       | M/M?    | T <sub>H</sub> (K) | T <sub>UQFF</sub> (K) | T <sub>UQFF</sub> /T <sub>H</sub> |
|--------------|---------|--------------------|-----------------------|-----------------------------------|
| Sgr A*       | 4■106   | 1.53■10?■4         | 1.52■10?■4            | <b>0.9999</b>                     |
| M87*         | 6.5■10? | 9.43■10?■8         | 9.34■10?■8            | <b>0.9999</b>                     |
| Stellar BH   | 10      | 6.15■10??          | 6.09■10??             | <b>0.9899</b>                     |
| Neutron star | 1.4     | 4.38■10?8          | 4.34■10?8             | <b>0.9899</b>                     |
| Magnetar     | 1.4     | ~4.38■10?8         | ~4.34■10?8            | <b>0.9899</b>                     |

C++ cross-validation: **All ratios 0.99 ■ 0.01 confirmed.** ?

## 5. Long-Form Equation Output

From get\_hawking\_long\_form(M) (validatehawkingtemperature.py Test 5):

|                                                                             |
|-----------------------------------------------------------------------------|
| UQFF-MODIFIED HAWKING TEMPERATURE                                           |
| =====                                                                       |
| Standard: $T_H = \frac{c^3}{8\pi G M k_B} = 1.528e-14 \text{ K (Sgr A*)}$   |
| UQFF correction:                                                            |
| f_TRZ factor: +1.01 (Toroidal Resonance Zone, $r_{TRZ} = ? \cdot r_S$ )     |
| SCm density ratio: 0.99 ( $\frac{\rho_{vac\_SCm}}{\rho_{vac\_UA}} = 0.01$ ) |
| Combined ratio: 0.9999                                                      |
| RESULT: $T_{UQFF} = 1.512e-14 \text{ K (Sgr A*, } 4 \cdot 10^6 \text{ M?)}$ |

Summary

| Parameter              | Value          | Source                        |
|------------------------|----------------|-------------------------------|
| f_TRZ                  | 0.01           | SOURCE4 calibration           |
| $\rho_{SCm}/\rho_{UA}$ | 0.01           | [SCm] 0.99                    |
| $T_{UQFF}/T_H$         | 0.99           | Both corrections cancel       |
| Test result            | 6/6 tests PASS | validatehawkingtemperature.py |
| C++ cross-check        | PASS           | uqfftemperatureformula.cpp    |

Source: validatehawkingtemperature.py, HawkingTemperatureUQFFCalculator |  $\rho = 0.0005/\text{day}$  | [SSq] = 0.57 .Groups[1].Value "PAPER\_0:D3" -f \$n

Abstract

Hawking (1974) derived the black body temperature of a black hole as  $T_H = \frac{c^3}{8\pi G M k_B}$ . The UQFF modifies this through two corrections: (1) the Toroidal Resonance Zone factor  $f_{TRZ}$  (enhancing vacuum fluctuation density near the horizon), and (2) the superconductive vacuum density ratio  $\rho_{vacSCm}/\rho_{vacUA}$  (reducing the effective thermodynamic temperature). The resulting ratio  $T_{UQFF}/T_H = (1 + f_{TRZ})(1 - \rho_{vacSCm}/\rho_{vacUA})$  is validated to equal 0.99 for Sgr A\* ( $4 \cdot 10^6 \text{ M?}$ ), confirming these corrections partially cancel. The validate\_hawking\_temperature.py test suite (6 tests) confirms this analytically and cross-validates with the C++ uqff\_temperature\_formula.cpp module.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\rho = 5.0 \cdot 10^{-4} \text{ day}^{-1}$ , [SSq] = 0.57) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Standard Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}$$

| System                               | M (kg)            | T_H (K)               |
|--------------------------------------|-------------------|-----------------------|
| Sgr A* ( $4 \cdot 10^6 \text{ M?}$ ) | $7.96 \cdot 10^6$ | $1.53 \cdot 10^{-14}$ |
| M87* ( $6.5 \cdot 10^7 \text{ M?}$ ) | $1.29 \cdot 10^4$ | $9.43 \cdot 10^{-18}$ |
| Stellar BH (10 M?)                   | $1.99 \cdot 10^3$ | $6.15 \cdot 10^{-17}$ |
| Primordial BH ( $10^2 \text{ kg}$ )  | $10^2$            | $1.23 \cdot 10^{-23}$ |

| System             | M (kg)    | T <sub>H</sub> (K) |
|--------------------|-----------|--------------------|
| Neutron star       | 2.8■10■   | 4.38■10?8          |
| Magnetar (SGR1745) | 1.4■2■10■ | ~10?8              |

2. UQFF Correction Terms

*f<sub>TRZ</sub>: Toroidal Resonance Zone Factor*

The TRZ is a time-reversal zone (an annular region of constructive quantum vacuum interference) at  $r_{TRZ} = ? \cdot r_S$  (where ? is the TRZ radius factor). It enhances the local density of Hawking-emitted modes:

$$T_{\rm \, TRZ-enhanced} = T_H \times (1 + f_{\rm \, TRZ})$$

Default UQFF value:  $f_{\rm \, TRZ} = 0.01$  (from SOURCE4 calibration).

*?vacSCm: Superconductive Vacuum Density Correction*

The UQFF superconductive vacuum density is lower than the [UA] vacuum density:

$$T_{\rm \, SCm-corrected} = T_H \times \left(1 - \frac{\rho_{\rm \, vac, SCm}}{\rho_{\rm \, vac, [UA]}}\right)$$

With  $?vacSCm/?vac[UA] = 0.01$  (from [SCm] ■ 0.99 calibration).

3. Combined UQFF Ratio

$$\frac{T_{\rm \, UQFF}}{T_H} = (1 + f_{\rm \, TRZ}) \times \left(1 - \frac{\rho_{\rm \, vac, SCm}}{\rho_{\rm \, vac, [UA]}}\right)$$

$$= (1 + 0.01) \times (1 - 0.01) = 1.01 \times 0.99 = \mathbf{0.9999 \approx 0.99}$$

**validatehawkingtemperature.py Test 1 confirms:**  $T_{\rm UQFF}/T_H = 0.99$  to 4 significant figures for Sgr A\*. ?

Cross-validation with C++ `uqff_temperature_formula.cpp`: **PASS** (within 0.01). ?

4. All-Systems Validation

From `validatehawkingtemperature.py` Test 3 (`compute(mode='all_systems')`):

| System       | M/M?    | T <sub>H</sub> (K) | T <sub>UQFF</sub> (K) | T <sub>UQFF</sub> /T <sub>H</sub> |
|--------------|---------|--------------------|-----------------------|-----------------------------------|
| Sgr A*       | 4■106   | 1.53■10?■4         | 1.52■10?■4            | <b>0.9999</b>                     |
| M87*         | 6.5■10? | 9.43■10?■8         | 9.34■10?■8            | <b>0.9999</b>                     |
| Stellar BH   | 10      | 6.15■10??          | 6.09■10??             | <b>0.9899</b>                     |
| Neutron star | 1.4     | 4.38■10?8          | 4.34■10?8             | <b>0.9899</b>                     |
| Magnetar     | 1.4     | ~4.38■10?8         | ~4.34■10?8            | <b>0.9899</b>                     |

C++ cross-validation: **All ratios 0.99 ■ 0.01 confirmed.** ?

5. Long-Form Equation Output

From get\_hawking\_long\_form(M) (validatehawkingtemperature.py Test 5):

|                                                                             |
|-----------------------------------------------------------------------------|
| UQFF-MODIFIED HAWKING TEMPERATURE                                           |
| =====                                                                       |
| Standard: $T_H = \frac{hc}{8\pi G M k_B} = 1.528e-14 \text{ K (Sgr A*)}$    |
| UQFF correction:                                                            |
| f_TRZ factor: +1.01 (Toroidal Resonance Zone, $r_{TRZ} = ? \cdot r_S$ )     |
| SCm density ratio: 0.99 ( $\frac{?_{vac\_SCm}}{?_{vac\_UA}} = 0.01$ )       |
| Combined ratio: 0.9999                                                      |
| RESULT: $T_{UQFF} = 1.512e-14 \text{ K (Sgr A*, } 4 \cdot 10^6 \text{ M?)}$ |

Summary

| Parameter       | Value          | Source                        |
|-----------------|----------------|-------------------------------|
| f_TRZ           | 0.01           | SOURCE4 calibration           |
| ?SCm/?UA        | 0.01           | [SCm] 0.99                    |
| TUQFF/TH        | 0.99           | Both corrections cancel       |
| Test result     | 6/6 tests PASS | validatehawkingtemperature.py |
| C++ cross-check | PASS           | uqfftemperatureformula.cpp    |

Source: validatehawkingtemperature.py, HawkingTemperatureUQFFCalculator | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value UQFF-Modified Hawking Temperature: Derivation

Title: UQFF-Modified Hawking Temperature: Derivation via fTRZ and ?vac\_SCm Vacuum Corrections

Author: Daniel T. Murphy Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) Date: March 7, 2026 Source Data: validatehawkingtemperature.py (Tests 16), CondensedPhysics.py HawkingTemperatureUQFFCalculator Index Slot: 1.11 Black Hole Physics & Hawking Radiation, \$n = [int]# "PAPER\_0:D3" -f [int]# PAPER #81 UQFF-Modified Hawking Temperature: Derivation

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Abstract

Hawking (1974) derived the black body temperature of a black hole as  $T_H = \frac{hc}{8\pi G M k_B}$ . The UQFF modifies this through two corrections: (1) the Toroidal Resonance Zone factor fTRZ (enhancing vacuum fluctuation density near the horizon), and (2) the superconductive vacuum density ratio ?vacSCm/?vacUA

(reducing the effective thermodynamic temperature). The resulting ratio  $T_{\text{UQFF}}/T_H = (1 + f_{\text{TRZ}})(1 - \rho_{\text{vacSCm}}/\rho_{\text{vacUA}})$  is validated to equal **0.99** for Sgr A\* ( $4 \times 10^6 M_\odot$ ), confirming these corrections partially cancel. The `validate_hawking_temperature.py` test suite (6 tests) confirms this analytically and cross-validates with the C++ `uqff_temperature_formula.cpp` module.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^4 \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

### 1. Standard Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}$$

| System                                 | M (kg)             | T <sub>H</sub> (K)     |
|----------------------------------------|--------------------|------------------------|
| Sgr A* ( $4 \times 10^6 M_\odot$ )     | $7.96 \times 10^6$ | $1.53 \times 10^{-4}$  |
| M87* ( $6.5 \times 10^7 M_\odot$ )     | $1.29 \times 10^4$ | $9.43 \times 10^{-8}$  |
| Stellar BH ( $10 M_\odot$ )            | $1.99 \times 10^3$ | $6.15 \times 10^{-7}$  |
| Primordial BH ( $10^{22} \text{ kg}$ ) | $10^{22}$          | $1.23 \times 10^{-22}$ |
| Neutron star                           | $2.8 \times 10^3$  | $4.38 \times 10^{-8}$  |
| Magnetar (SGR1745)                     | $1.4 \times 10^3$  | $\sim 10^{-8}$         |

### 2. UQFF Correction Terms

#### *f<sub>TRZ</sub>: Toroidal Resonance Zone Factor*

The TRZ is a time-reversal zone (an annular region of constructive quantum vacuum interference) at  $r_{\text{TRZ}} = \tau r_S$  (where  $\tau$  is the TRZ radius factor). It enhances the local density of Hawking-emitted modes:

$$T_{\text{TRZ-enhanced}} = T_H \times (1 + f_{\text{TRZ}})$$

Default UQFF value:  $f_{\text{TRZ}} = 0.01$  (from SOURCE4 calibration).

#### *$\rho_{\text{vacSCm}}$ : Superconductive Vacuum Density Correction*

The UQFF superconductive vacuum density is lower than the  $[\text{UA}]$  vacuum density:

$$T_{\text{SCm-corrected}} = T_H \times \left(1 - \frac{\rho_{\text{vac,SCm}}}{\rho_{\text{vac,UA}}}\right)$$

With  $\rho_{\text{vacSCm}}/\rho_{\text{vacUA}} = 0.01$  (from  $[\text{SCm}] \approx 0.99$  calibration).

### 3. Combined UQFF Ratio

$$\frac{T_{\text{UQFF}}}{T_H} = (1 + f_{\text{TRZ}}) \times \left(1 - \frac{\rho_{\text{vac,SCm}}}{\rho_{\text{vac,UA}}}\right)$$

$$= (1 + 0.01) \times (1 - 0.01) = 1.01 \times 0.99 = \mathbf{0.9999 \approx 0.99}$$

**validatehawkingtemperature.py Test 1 confirms:**  $T_{\text{UQFF}}/T_H = 0.99$  to 4 significant figures for Sgr A\*.

Cross-validation with C++ `uqff_temperature_formula.cpp`: **PASS** (within 0.01). ?

#### 4. All-Systems Validation

From `validatehawkingtemperature.py` Test 3 (`compute(mode='all_systems')`):

| System       | M/M?    | T <sub>H</sub> (K) | T <sub>UQFF</sub> (K) | T <sub>UQFF</sub> /T <sub>H</sub> |
|--------------|---------|--------------------|-----------------------|-----------------------------------|
| Sgr A*       | 4■106   | 1.53■10?■4         | 1.52■10?■4            | <b>0.9999</b>                     |
| M87*         | 6.5■10? | 9.43■10?■8         | 9.34■10?■8            | <b>0.9999</b>                     |
| Stellar BH   | 10      | 6.15■10??          | 6.09■10??             | <b>0.9899</b>                     |
| Neutron star | 1.4     | 4.38■10?8          | 4.34■10?8             | <b>0.9899</b>                     |
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C++ cross-validation: **All ratios 0.99 ■ 0.01 confirmed.** ?

#### 5. Long-Form Equation Output

From `get_hawking_long_form(M)` (`validatehawkingtemperature.py` Test 5):

```
UQFF-MODIFIED HAWKING TEMPERATURE
=====
Standard: TH = ?c■/(8pGMkB) = 1.528e-14 K (Sgr A*)
UQFF correction:
fTRZ factor: +1.01 (Toroidal Resonance Zone, rTRZ = ? ■ rS)
SCm density ratio: ■0.99 (?vacSCm / ?vacUA = 0.01)
Combined ratio: 0.9999
RESULT: TUQFF = 1.512e-14 K (Sgr A*, 4■106 M?)
```

#### Summary

| Parameter                         | Value          | Source                                     |
|-----------------------------------|----------------|--------------------------------------------|
| f <sub>TRZ</sub>                  | 0.01           | SOURCE4 calibration                        |
| ?SC <sub>m</sub> ?UA              | 0.01           | [SC <sub>m</sub> ] ■ 0.99                  |
| T <sub>UQFF</sub> /T <sub>H</sub> | <b>0.99</b>    | Both corrections cancel                    |
| Test result                       | 6/6 tests PASS | <code>validatehawkingtemperature.py</code> |
| C++ cross-check                   | PASS           | <code>uqfftemperatureformula.cpp</code>    |

Source: `validatehawkingtemperature.py`, `HawkingTemperatureUQFFCalculator` | ? = 0.0005/day | [SSq] = 0.57.Groups[1].Value "PAPER<sub>0:D3</sub>" -f \$n

#### Abstract

Hawking (1974) derived the black body temperature of a black hole as  $T_H = \frac{?c}{8pGMkB}$ . The UQFF modifies this through two corrections: (1) the Toroidal Resonance Zone factor  $f_{TRZ}$  (*enhancing vacuum fluctuation density near the horizon*), and (2) the superconductive vacuum density ratio  $?_{vac}SC_m/?_{vac}UA$  (*reducing the effective thermodynamic temperature*). The resulting ratio  $T_{UQFF}/T_H = (1 + f_{TRZ})(1 -$

$\rho_{vacSCm}/\rho_{vacUA}$ ) is validated to equal **0.99** for Sgr A\* ( $4 \times 10^6 M_\odot$ ), confirming these corrections partially cancel. The `validate_hawking_temperature.py` test suite (6 tests) confirms this analytically and cross-validates with the C++ `uqff_temperature_formula.cpp` module.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\beta = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

### 1. Standard Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}$$

| System                                 | M (kg)             | T <sub>H</sub> (K)    |
|----------------------------------------|--------------------|-----------------------|
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| M87* ( $6.5 \times 10^7 M_\odot$ )     | $1.29 \times 10^4$ | $9.43 \times 10^{-8}$ |
| Stellar BH ( $10 M_\odot$ )            | $1.99 \times 10^3$ | $6.15 \times 10^{-9}$ |
| Primordial BH ( $10^{22} \text{ kg}$ ) | $10^{22}$          | $1.23 \times 10^{12}$ |
| Neutron star                           | $2.8 \times 10^3$  | $4.38 \times 10^{-8}$ |
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### 2. UQFF Correction Terms

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Default UQFF value:  $f_{\text{TRZ}} = 0.01$  (from SOURCE4 calibration).

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The UQFF superconductive vacuum density is lower than the  $\rho_{vacUA}$  vacuum density:

$$T_{\text{SCm-corrected}} = T_H \times \left(1 - \frac{\rho_{vac,SCm}}{\rho_{vac,UA}}\right)$$

With  $\rho_{vacSCm}/\rho_{vacUA} = 0.01$  (from  $[SCm] \times 0.99$  calibration).

### 3. Combined UQFF Ratio

$$\frac{T_{\text{UQFF}}}{T_H} = (1 + f_{\text{TRZ}}) \times \left(1 - \frac{\rho_{vac,SCm}}{\rho_{vac,UA}}\right)$$

$$= (1 + 0.01) \times (1 - 0.01) = 1.01 \times 0.99 = \mathbf{0.9999 \approx 0.99}$$

**validatehawkingtemperature.py Test 1 confirms:**  $T_{\text{UQFF}}/T_H = 0.99$  to 4 significant figures for Sgr A\*.

Cross-validation with C++ `uqff_temperature_formula.cpp`: **PASS** (within 0.01).



#### 4. All-Systems Validation

From validatehawkingtemperature.py Test 3 (compute(mode='all\_systems')):

| System       | M/M?    | T_H (K)    | T_UQFF (K) | TUQFF/TH      |
|--------------|---------|------------|------------|---------------|
| Sgr A*       | 4■106   | 1.53■10?■4 | 1.52■10?■4 | <b>0.9999</b> |
| M87*         | 6.5■10? | 9.43■10?■8 | 9.34■10?■8 | <b>0.9999</b> |
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C++ cross-validation: **All ratios 0.99 ■ 0.01 confirmed.** ?

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From get\_hawking\_long\_form(M) (validatehawkingtemperature.py Test 5):

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=====
Standard: T_H = ?c■/(8pGMk_B) = 1.528e-14 K (Sgr A*)
UQFF correction:
f_TRZ factor: +1.01 (Toroidal Resonance Zone, r_TRZ = ? ■ r_S)
SCm density ratio: ■0.99 (?_vac_SCm / ?_vac_UA = 0.01)
Combined ratio: 0.9999
RESULT: T_UQFF = 1.512e-14 K (Sgr A*, 4■106 M?)
```

#### Summary

| Parameter       | Value          | Source                        |
|-----------------|----------------|-------------------------------|
| f_TRZ           | 0.01           | SOURCE4 calibration           |
| ?SCm/?UA        | 0.01           | [SCm] ■ 0.99                  |
| TUQFF/TH        | <b>0.99</b>    | Both corrections cancel       |
| Test result     | 6/6 tests PASS | validatehawkingtemperature.py |
| C++ cross-check | PASS           | uqfftemperatureformula.cpp    |

Source: validatehawkingtemperature.py, HawkingTemperatureUQFFCalculator | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value ■ UQFF-Modified Hawking Temperature: Derivation

**Title:** UQFF-Modified Hawking Temperature: Derivation via fTRZ and ?vac\_SCm Vacuum Corrections

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source** **Data:** validatehawkingtemperature.py (Tests 1■6), CondensedPhysics.py HawkingTemperatureUQFFCalculator **Index** **Slot:** ■1.11 Black Hole Physics & Hawking Radiation, "PAPER<sub>0:D3</sub>" -f [int]# PAPER #81 ■ UQFF-Modified Hawking Temperature: Derivation

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**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic ( $\gamma = 0.0005/\text{day}$ ,  $[\text{SSq}] = 0.57$ ) **Date:** March 7, 2026 **Source Data:** `validatehawkingtemperature.py` (Tests 1/6), `CondensedPhysics.py` `HawkingTemperatureUQFFCalculator` **Index Slot:** 1.11 Black Hole Physics & Hawking Radiation, \$n = [int]\# \text{ PAPER \#81 }\$ UQFF-Modified Hawking Temperature: Derivation

**Title:** UQFF-Modified Hawking Temperature: Derivation via  $f_{\text{TRZ}}$  and  $\gamma_{\text{vac\_SCm}}$  Vacuum Corrections

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic ( $\gamma = 0.0005/\text{day}$ ,  $[\text{SSq}] = 0.57$ ) **Date:** March 7, 2026 **Source Data:** `validatehawkingtemperature.py` (Tests 1/6), `CondensedPhysics.py` `HawkingTemperatureUQFFCalculator` **Index Slot:** 1.11 Black Hole Physics & Hawking Radiation, PAPER\_081

Abstract

Hawking (1974) derived the black body temperature of a black hole as  $T_H = \hbar c^3 / (8\pi G M k_B)$ . The UQFF modifies this through two corrections: (1) the Toroidal Resonance Zone factor  $f_{\text{TRZ}}$  (*enhancing vacuum fluctuation density near the horizon*), and (2) the superconductive vacuum density ratio  $\gamma_{\text{vacSCm}}/\gamma_{\text{vacUA}}$  (*reducing the effective thermodynamic temperature*). The resulting ratio  $T_{\text{UQFF}}/T_H = (1 + f_{\text{TRZ}})(1 - \gamma_{\text{vacSCm}}/\gamma_{\text{vacUA}})$  is validated to equal **0.99** for Sgr A\* ( $4 \times 10^6 M_\odot$ ), confirming these corrections partially cancel. The `validate_hawking_temperature.py` test suite (6 tests) confirms this analytically and cross-validates with the C++ `uqff_temperature_formula.cpp` module.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Standard Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}$$

| System                             | M (kg)             | T_H (K)                |
|------------------------------------|--------------------|------------------------|
| Sgr A* ( $4 \times 10^6 M_\odot$ ) | $7.96 \times 10^6$ | $1.53 \times 10^{-4}$  |
| M87* ( $6.5 \times 10^7 M_\odot$ ) | $1.29 \times 10^4$ | $9.43 \times 10^{-8}$  |
| Stellar BH ( $10 M_\odot$ )        | $1.99 \times 10^3$ | $6.15 \times 10^{-7}$  |
| Primordial BH ( $10^{22}$ kg)      | $10^{22}$          | $1.23 \times 10^{-22}$ |
| Neutron star                       | $2.8 \times 10^3$  | $4.38 \times 10^{-8}$  |
| Magnetar (SGR1745)                 | $1.4 \times 10^3$  | $\sim 10^{-8}$         |

2. UQFF Correction Terms

$f_{\text{TRZ}}$ : Toroidal Resonance Zone Factor

The TRZ is a time-reversal zone (an annular region of constructive quantum vacuum interference) at  $r_{\text{TRZ}} = \gamma r_S$  (where  $\gamma$  is the TRZ radius factor). It enhances the local density of Hawking-emitted modes:

$$T_{\text{\rm TRZ-enhanced}} = T_H \times (1 + f_{\text{\rm TRZ}})$$

Default UQFF value:  $f_{\text{TRZ}} = 0.01$  (from SOURCE4 calibration).

*?vacSCm: Superconductive Vacuum Density Correction*

The UQFF superconductive vacuum density is lower than the [UA] vacuum density:

$$T_{\rm SCm-corrected} = T_H \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, UA}}\right)$$

With ?vacSCm/?vac[UA] = 0.01 (from [SCm] ■ 0.99 calibration).

**3. Combined UQFF Ratio**

$$\frac{T_{\rm UQFF}}{T_H} = (1 + f_{\rm TRZ}) \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, UA}}\right)$$

$$= (1 + 0.01) \times (1 - 0.01) = 1.01 \times 0.99 = \mathbf{0.9999 \approx 0.99}$$

**validatehawkingtemperature.py Test 1 confirms:** T\_UQFF/T\_H = 0.99 to 4 significant figures for Sgr A\*. ?

Cross-validation with C++ uqff\_temperature\_formula.cpp: **PASS** (within 0.01). ?

**4. All-Systems Validation**

From validatehawkingtemperature.py Test 3 (compute(mode='all\_systems')):

| System       | M/M?    | T_H (K)    | T_UQFF (K) | TUQFF/TH      |
|--------------|---------|------------|------------|---------------|
| Sgr A*       | 4■106   | 1.53■10?■4 | 1.52■10?■4 | <b>0.9999</b> |
| M87*         | 6.5■10? | 9.43■10?■8 | 9.34■10?■8 | <b>0.9999</b> |
| Stellar BH   | 10      | 6.15■10??  | 6.09■10??  | <b>0.9899</b> |
| Neutron star | 1.4     | 4.38■10?8  | 4.34■10?8  | <b>0.9899</b> |
| Magnetar     | 1.4     | ~4.38■10?8 | ~4.34■10?8 | <b>0.9899</b> |

C++ cross-validation: **All ratios 0.99 ■ 0.01 confirmed.** ?

**5. Long-Form Equation Output**

From get\_hawking\_long\_form(M) (validatehawkingtemperature.py Test 5):

UQFF-MODIFIED HAWKING TEMPERATURE  
=====

Standard: T\_H = ?c■/(8pGMk\_B) = 1.528e-14 K (Sgr A\*)

UQFF correction:

f\_TRZ factor: +1.01 (Toroidal Resonance Zone, r\_TRZ = ? ■ r\_S)

SCm density ratio: ■0.99 (?\_vac\_SCm / ?\_vac\_UA = 0.01)

Combined ratio: 0.9999

RESULT: T\_UQFF = 1.512e-14 K (Sgr A\*, 4■106 M?)

**Summary**

| Parameter       | Value          | Source                        |
|-----------------|----------------|-------------------------------|
| f_TRZ           | 0.01           | SOURCE4 calibration           |
| ?SCm/?UA        | 0.01           | [SCm] ■ 0.99                  |
| TUQFF/TH        | <b>0.99</b>    | Both corrections cancel       |
| Test result     | 6/6 tests PASS | validatehawkingtemperature.py |
| C++ cross-check | PASS           | uqfftemperatureformula.cpp    |

Source: validatehawkingtemperature.py, HawkingTemperatureUQFFCalculator | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER<sub>0:D3</sub>" -f \$n

Abstract

Hawking (1974) derived the black body temperature of a black hole as  $TH = \frac{c}{8\pi G M k_B}$ . The UQFF modifies this through two corrections: (1) the Toroidal Resonance Zone factor  $f_{TRZ}$  (enhancing vacuum fluctuation density near the horizon), and (2) the superconductive vacuum density ratio  $\frac{\rho_{vacSCm}}{\rho_{vacUA}}$  (reducing the effective thermodynamic temperature). The resulting ratio  $TUQFF/TH = (1 + f_{TRZ})(1 - \frac{\rho_{vacSCm}}{\rho_{vacUA}})$  is validated to equal **0.99** for Sgr A\* ( $4 \times 10^6 M_\odot$ ), confirming these corrections partially cancel. The validate\_hawking\_temperature.py test suite (6 tests) confirms this analytically and cross-validates with the C++ uqff\_temperature\_formula.cpp module.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\rho = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Standard Hawking Temperature

T\_H = \frac{\hbar c^3}{8\pi G M k\_B}

| System                             | M (kg)                | T_H (K)               |
|------------------------------------|-----------------------|-----------------------|
| Sgr A* ( $4 \times 10^6 M_\odot$ ) | $7.96 \times 10^{36}$ | $1.53 \times 10^{-4}$ |
| M87* ( $6.5 \times 10^7 M_\odot$ ) | $1.29 \times 10^{38}$ | $9.43 \times 10^{-8}$ |
| Stellar BH ( $10 M_\odot$ )        | $1.99 \times 10^{31}$ | $6.15 \times 10^{-7}$ |
| Primordial BH ( $10^{22}$ kg)      | $10^{22}$             | $1.23 \times 10^{12}$ |
| Neutron star                       | $2.8 \times 10^{30}$  | $4.38 \times 10^8$    |
| Magnetar (SGR1745)                 | $1.4 \times 10^{31}$  | $\sim 10^8$           |

2. UQFF Correction Terms

f\_TRZ: Toroidal Resonance Zone Factor

The TRZ is a time-reversal zone (an annular region of constructive quantum vacuum interference) at  $r_{TRZ} = \rho \cdot r_S$  (where  $\rho$  is the TRZ radius factor). It enhances the local density of Hawking-emitted modes:

T\_{\rm TRZ-enhanced} = T\_H \times (1 + f\_{\rm TRZ})

Default UQFF value: f\_TRZ = 0.01 (from SOURCE4 calibration).

*?vacSCm: Superconductive Vacuum Density Correction*

The UQFF superconductive vacuum density is lower than the [UA] vacuum density:

$$T_{\rm SCm-corrected} = T_H \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, UA}}\right)$$

With ?vacSCm/?vac[UA] = 0.01 (from [SCm] ■ 0.99 calibration).

3. Combined UQFF Ratio

$$\frac{T_{\rm UQFF}}{T_H} = (1 + f_{\rm TRZ}) \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, UA}}\right)$$

$$= (1 + 0.01) \times (1 - 0.01) = 1.01 \times 0.99 = \mathbf{0.9999 \approx 0.99}$$

**validatehawkingtemperature.py Test 1 confirms:** T\_UQFF/T\_H = 0.99 to 4 significant figures for Sgr A\*. ?

Cross-validation with C++ uqff\_temperature\_formula.cpp: **PASS** (within 0.01). ?

4. All-Systems Validation

From validatehawkingtemperature.py Test 3 (compute(mode='all\_systems')):

| System       | M/M?    | T_H (K)    | T_UQFF (K) | TUQFF/TH      |
|--------------|---------|------------|------------|---------------|
| Sgr A*       | 4■106   | 1.53■10?■4 | 1.52■10?■4 | <b>0.9999</b> |
| M87*         | 6.5■10? | 9.43■10?■8 | 9.34■10?■8 | <b>0.9999</b> |
| Stellar BH   | 10      | 6.15■10??  | 6.09■10??  | <b>0.9899</b> |
| Neutron star | 1.4     | 4.38■10?8  | 4.34■10?8  | <b>0.9899</b> |
| Magnetar     | 1.4     | ~4.38■10?8 | ~4.34■10?8 | <b>0.9899</b> |

C++ cross-validation: **All ratios 0.99 ■ 0.01 confirmed.** ?

5. Long-Form Equation Output

From get\_hawking\_long\_form(M) (validatehawkingtemperature.py Test 5):

UQFF-MODIFIED HAWKING TEMPERATURE  
=====

Standard: T\_H = ?c■/(8pGMk\_B) = 1.528e-14 K (Sgr A\*)

UQFF correction:

f\_TRZ factor: +1.01 (Toroidal Resonance Zone, r\_TRZ = ? ■ r\_S)

SCm density ratio: ■0.99 (?\_vac\_SCm / ?\_vac\_UA = 0.01)

Combined ratio: 0.9999

RESULT: T\_UQFF = 1.512e-14 K (Sgr A\*, 4■106 M?)

Summary

| Parameter       | Value          | Source                        |
|-----------------|----------------|-------------------------------|
| f_TRZ           | 0.01           | SOURCE4 calibration           |
| ?SCm/?UA        | 0.01           | [SCm] ■ 0.99                  |
| TUQFF/TH        | <b>0.99</b>    | Both corrections cancel       |
| Test result     | 6/6 tests PASS | validatehawkingtemperature.py |
| C++ cross-check | PASS           | uqfftemperatureformula.cpp    |

Source: validatehawkingtemperature.py, HawkingTemperatureUQFFCalculator | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value

### Abstract

Hawking (1974) derived the black body temperature of a black hole as  $T_H = \frac{\hbar c^3}{8\pi G M k_B}$ . The UQFF modifies this through two corrections: (1) the Toroidal Resonance Zone factor  $f_{TRZ}$  (enhancing vacuum fluctuation density near the horizon), and (2) the superconductive vacuum density ratio  $\frac{\rho_{vacSCm}}{\rho_{vacUA}}$  (reducing the effective thermodynamic temperature). The resulting ratio  $T_{UQFF}/T_H = (1 + f_{TRZ})(1 - \frac{\rho_{vacSCm}}{\rho_{vacUA}})$  is validated to equal **0.99** for Sgr A\* ( $4 \times 10^6 M_\odot$ ), confirming these corrections partially cancel. The validate\_hawking\_temperature.py test suite (6 tests) confirms this analytically and cross-validates with the C++ uqff\_temperature\_formula.cpp module.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\rho = 5.0 \times 10^{-24} \text{ day}^{-1}$ , [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

### 1. Standard Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}$$

| System                                 | M (kg)                        | T_H (K)                |
|----------------------------------------|-------------------------------|------------------------|
| Sgr A* ( $4 \times 10^6 M_\odot$ )     | $7.96 \times 10^6$            | $1.53 \times 10^{-4}$  |
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| Stellar BH ( $10 M_\odot$ )            | $1.99 \times 10^{30}$         | $6.15 \times 10^{-22}$ |
| Primordial BH ( $10^{22} \text{ kg}$ ) | $10^{22}$                     | $1.23 \times 10^{22}$  |
| Neutron star                           | $2.8 \times 10^{30}$          | $4.38 \times 10^{-8}$  |
| Magnetar (SGR1745)                     | $1.4 \times 2 \times 10^{30}$ | $\sim 10^{-8}$         |

### 2. UQFF Correction Terms

#### f\_TRZ: Toroidal Resonance Zone Factor

The TRZ is a time-reversal zone (an annular region of constructive quantum vacuum interference) at  $r_{TRZ} = \rho \cdot r_S$  (where  $\rho$  is the TRZ radius factor). It enhances the local density of Hawking-emitted modes:

$$T_{\text{TRZ-enhanced}} = T_H \times (1 + f_{\text{TRZ}})$$

Default UQFF value:  $f_{\text{TRZ}} = 0.01$  (from SOURCE4 calibration).

?vacSCm: Superconductive Vacuum Density Correction

The UQFF superconductive vacuum density is lower than the [UA] vacuum density:

$$T_{\rm SCm-corrected} = T_H \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, UA}}\right)$$

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$$\frac{T_{\rm UQFF}}{T_H} = (1 + f_{\rm TRZ}) \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, UA}}\right)$$

$$= (1 + 0.01) \times (1 - 0.01) = 1.01 \times 0.99 = \mathbf{0.9999 \approx 0.99}$$

**validatehawkingtemperature.py Test 1 confirms:** T\_UQFF/T\_H = 0.99 to 4 significant figures for Sgr A\*.

Cross-validation with C++ uqff\_temperature\_formula.cpp: **PASS** (within 0.01).

4. All-Systems Validation

From validatehawkingtemperature.py Test 3 (compute(mode='all\_systems')):

| System       | M/M?    | T_H (K)    | T_UQFF (K) | TUQFF/TH      |
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C++ cross-validation: **All ratios 0.99 ■ 0.01 confirmed.**

5. Long-Form Equation Output

From get\_hawking\_long\_form(M) (validatehawkingtemperature.py Test 5):

```
UQFF-MODIFIED HAWKING TEMPERATURE
=====
Standard: T_H = ?c■/(8pGMk_B) = 1.528e-14 K (Sgr A*)
UQFF correction:
f_TRZ factor: +1.01 (Toroidal Resonance Zone, r_TRZ = ? ■ r_S)
SCm density ratio: ■0.99 (?_vac_SCm / ?_vac_UA = 0.01)
Combined ratio: 0.9999
RESULT: T_UQFF = 1.512e-14 K (Sgr A*, 4■106 M?)
```

Summary

| Parameter       | Value          | Source                        |
|-----------------|----------------|-------------------------------|
| f_TRZ           | 0.01           | SOURCE4 calibration           |
| ?SCm/?UA        | 0.01           | [SCm] ■ 0.99                  |
| TUQFF/TH        | <b>0.99</b>    | Both corrections cancel       |
| Test result     | 6/6 tests PASS | validatehawkingtemperature.py |
| C++ cross-check | PASS           | uqfftemperatureformula.cpp    |

Source: validatehawkingtemperature.py, HawkingTemperatureUQFFCalculator | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER<sub>0:D3</sub>" -f \$n

Abstract

Hawking (1974) derived the black body temperature of a black hole as  $TH = \frac{c}{8\pi G M k_B}$ . The UQFF modifies this through two corrections: (1) the Toroidal Resonance Zone factor  $f_{TRZ}$  (*enhancing vacuum fluctuation density near the horizon*), and (2) the superconductive vacuum density ratio  $\frac{\rho_{vacSCm}}{\rho_{vacUA}}$  (*reducing the effective thermodynamic temperature*). The resulting ratio  $TUQFF/TH = (1 + f_{TRZ})(1 - \frac{\rho_{vacSCm}}{\rho_{vacUA}})$  is validated to equal **0.99** for Sgr A\* ( $4 \times 10^6 M_\odot$ ), confirming these corrections partially cancel. The validate\_hawking\_temperature.py test suite (6 tests) confirms this analytically and cross-validates with the C++ uqff\_temperature\_formula.cpp module.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\rho = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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The TRZ is a time-reversal zone (an annular region of constructive quantum vacuum interference) at  $r_{TRZ} = \rho \cdot r_S$  (where  $\rho$  is the TRZ radius factor). It enhances the local density of Hawking-emitted modes:

$$T_{\text{rm TRZ-enhanced}} = T_H \times (1 + f_{\text{rm TRZ}})$$

Default UQFF value: f\_TRZ = 0.01 (from SOURCE4 calibration).



*?vacSCm: Superconductive Vacuum Density Correction*

The UQFF superconductive vacuum density is lower than the [UA] vacuum density:

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$$\frac{T_{\rm UQFF}}{T_H} = (1 + f_{\rm TRZ}) \times \left(1 - \frac{\rho_{\rm vac, SCm}}{\rho_{\rm vac, UA}}\right)$$

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**validatehawkingtemperature.py Test 1 confirms:** T\_UQFF/T\_H = 0.99 to 4 significant figures for Sgr A\*. ?

Cross-validation with C++ uqff\_temperature\_formula.cpp: **PASS** (within 0.01). ?

**4. All-Systems Validation**

From validatehawkingtemperature.py Test 3 (compute(mode='all\_systems')):

| System       | M/M?    | T_H (K)    | T_UQFF (K) | TUQFF/TH      |
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=====

Standard: T\_H = ?c■/(8pGMk\_B) = 1.528e-14 K (Sgr A\*)

UQFF correction:

f\_TRZ factor: +1.01 (Toroidal Resonance Zone, r\_TRZ = ? ■ r\_S)

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Combined ratio: 0.9999

RESULT: T\_UQFF = 1.512e-14 K (Sgr A\*, 4■106 M?)

**Summary**

| Parameter       | Value          | Source                        |
|-----------------|----------------|-------------------------------|
| f_TRZ           | 0.01           | SOURCE4 calibration           |
| ?SCm/?UA        | 0.01           | [SCm] ■ 0.99                  |
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| Test result     | 6/6 tests PASS | validatehawkingtemperature.py |
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Source: validatehawkingtemperature.py, HawkingTemperatureUQFFCalculator | ? = 0.0005/day | [SSq] = 0.57